

stats

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Packages

```
library(tidyr)
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
library(ggsignif)
library(gtsummary)
library(dplyr)
library(rstatix)
library(ggpp)
library(lmerTest)
library(broom)
library(broom.mixed)
library(purrr)
library(gt)
```

Data

```
# Data Frame with ALL sites, TC (utc and tc isco's), CC, IT
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
```

```

      delta_d = processed_delta_d,
      delta_d_std = processed_delta_d_stddev,
      delta_o = processed_delta_18o,
      delta_o_std = processed_delta_18o_stddev) |>
mutate(date = mdy(date)) # telling r the date format

river_reach <- compiled_isotope |>
  filter(!site %in% c("utc_isco", "tc_isco")) |>
  mutate(site = factor(site, levels = c(
    "tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3",
    "coal_1", "coal_2", "coal_3", "coal_4", "coal_5", "coal_6",
    "italian_1", "italian_2", "italian_3", "italian_4", "italian_5"
  ))) |>
  arrange(site)

# Trail Creek site order
site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

# Trail Creek data
trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

# Coal Creek data
coal_creek <- compiled_isotope |>
  filter(location %in% c("cc"))

# Italian Creek data
italian_creek <- compiled_isotope |>
  filter(location == "it")

# Trail Creek ISCO data combined
trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_stddev,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

```

```

# Upstream Trail Creek ISCO
utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

# Downstream Trail Creek ISCO
tc_isco <- trail_creek_isco |>
  filter(site == "tc_isco")

# Trail Creek gradient data
gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

# Coal Creek gradient data
gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

# Italian Creek gradient data
gradient_it <- read_csv(here("data", "it_gradient_data.csv"))

# ALL Discharge data
discharge_isco <- read_csv(here("data", "isco_discharge_lc.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    period = if_else(date < as.Date("2026-05-14"), "pre-peak", "post-peak") # before & after
  )

# Upstream Trail Creek ISCO
discharge_utc_isco <- discharge_isco |>
  filter(isco %in% c("utc"))

# Downstream Trail Creek ISCO
discharge_tc_isco <- discharge_isco |>
  filter(isco %in% c("tc"))

# Upstream and Downstream Trail Creek ISCO
us_ds <- read_csv(here("data", "us_ds.csv")) |>
  mutate(
    date = as.Date(date, format = "%m/%d/%Y"),
    creek = case_when(
      grepl("^cc", location) ~ "Coal Creek",
      grepl("^it", location) ~ "Italian Creek",
      location %in% c("utc", "tc") ~ "Trail Creek"),
    reach = case_when(
      location %in% c("cc-us", "it-us", "utc") ~ "upstream",

```

```

    location %in% c("cc-ds", "it-ds", "tc") ~ "downstream"))

# Summary stats per period (per reach, if you have upstream/downstream)
discharge_stat <- discharge_isco |>
  group_by(period) |> # add reach here: group_by(reach, period)
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE)
  )

sontek <- read_csv(here("data", "sontek-cc-it.csv"), show_col_types = FALSE) |>
  clean_names() |>
  filter(!is.na(site)) |>
  separate(site, into = c("creek", "site_num"), sep = "-", remove = FALSE, convert = TRUE)

```

Statistical Tests

```

# ANOVA,
# lc-excess and location
anova <- aov(lc_excess ~ location, data = compiled_isotope)
summary(anova)

```

```

              Df Sum Sq Mean Sq F value Pr(>F)
location        3   209.2    69.72   36.53 <2e-16 ***
Residuals     111   211.8     1.91
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# TukeyHSD, compares every possible pair of group means
TukeyHSD(anova)

```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = lc_excess ~ location, data = compiled_isotope)
```

```
$location
```

	diff	lwr	upr	p adj
it-cc	2.2952900	0.8997570	3.6908230	0.0002219
tc-cc	-1.8984083	-3.0613525	-0.7354642	0.0002508
utc-cc	-0.7528817	-1.9388567	0.4330933	0.3520168
tc-it	-4.1936983	-5.2595543	-3.1278424	0.0000000
utc-it	-3.0481717	-4.1391100	-1.9572333	0.0000000
utc-tc	1.1455267	0.3741168	1.9169366	0.0010318

```
# Trail Creek
# lc-excess & normalized stream distance

# 6/24 - lc-excess & normalized stream distance
model.1 <- lm(`lc-excess_6-24` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.1)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

1	2	3	4	5	6	7
-0.36861	-0.32358	0.58444	0.03641	0.15093	0.31789	-0.39748

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.3907116	0.2562773	-5.427	0.002880 **
normalized_stream_distance_m	-0.0011403	0.0001537	-7.417	0.000702 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.416 on 5 degrees of freedom

Multiple R-squared: 0.9167, Adjusted R-squared: 0.9

F-statistic: 55.01 on 1 and 5 DF, p-value: 0.0007016

```
# 6/30 - lc-excess & normalized stream distance
model.2 <- lm(`lc-excess_6-30` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.2)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

	1	2	3	4	5	6	7
	-0.08528	-0.20669	0.24371	0.25137	0.19568	-0.61753	0.21874

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.2800413	0.2204137	-10.34	0.000145 ***
normalized_stream_distance_m	-0.0008555	0.0001322	-6.47	0.001314 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3578 on 5 degrees of freedom

Multiple R-squared: 0.8933, Adjusted R-squared: 0.872

F-statistic: 41.86 on 1 and 5 DF, p-value: 0.001314

```
# 7/15 - lc-excess & normalized stream distance
model.3 <- lm(`lc-excess_7-15` ~ normalized_stream_distance_m, data = gradient_tc)
summary(model.3)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ normalized_stream_distance_m,
    data = gradient_tc)
```

Residuals:

	1	2	3	4	5	6	7
	-0.36655	-0.34573	0.61325	0.47415	-0.07123	-0.28328	-0.02060

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.1034551	0.2671852	-4.130	0.00909 **
normalized_stream_distance_m	-0.0010277	0.0001603	-6.412	0.00137 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4337 on 5 degrees of freedom

Multiple R-squared: 0.8916, Adjusted R-squared: 0.8699

F-statistic: 41.11 on 1 and 5 DF, p-value: 0.001369

```
# Coal Creek
# lc-excess & normalized stream distance

# 6/23 - lc-excess & normalized stream distance
model.4 <- lm(`lc-excess_6-23` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.4)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

1	2	3	4	5
-0.3323	0.6279	0.2566	-0.3412	-0.2109

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.5393266	0.4492095	3.427	0.0416 *
normalized_stream_distance_m	-0.0006445	0.0002206	-2.921	0.0614 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4938 on 3 degrees of freedom

Multiple R-squared: 0.7399, Adjusted R-squared: 0.6532

F-statistic: 8.533 on 1 and 3 DF, p-value: 0.06144

```
# 7/7 - lc-excess & normalized stream distance
model.5 <- lm(`lc-excess_7-7` ~ normalized_stream_distance_m, data = gradient_cc)
summary(model.5)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ normalized_stream_distance_m,
    data = gradient_cc)
```

Residuals:

1	2	3	4	5
0.2092	0.3569	-0.8765	-0.8080	1.1184

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.333192	0.885667	-0.376	0.7318
normalized_stream_distance_m	-0.001031	0.000435	-2.369	0.0986

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9735 on 3 degrees of freedom

Multiple R-squared: 0.6517, Adjusted R-squared: 0.5356

F-statistic: 5.613 on 1 and 3 DF, p-value: 0.09859

```
# Italian Creek
# lc-excess & normalized stream distance

# 7/1 - lc-excess & normalized stream distance

model.6 <- lm(`lc_excess_7_1` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.6)
```

Call:

lm(formula = lc_excess_7_1 ~ normalized_stream_distance_m, data = gradient_it)

Residuals:

1	2	3	4	5
0.190779	-0.256938	-0.069233	0.141983	-0.006591

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.549e+00	1.634e-01	21.715	0.000214 ***
normalized_stream_distance_m	-9.279e-05	8.643e-05	-1.074	0.361692

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2061 on 3 degrees of freedom

Multiple R-squared: 0.2775, Adjusted R-squared: 0.03673

F-statistic: 1.153 on 1 and 3 DF, p-value: 0.3617

```
# 7/16 - lc-excess & normalized stream distance

model.7 <- lm(`lc_excess_7_16` ~ normalized_stream_distance_m, data = gradient_it)
summary(model.7)
```


Call:

```
lm(formula = `lc-excess_7_16` ~ normalized_stream_distance_m,  
    data = gradient_it)
```

Residuals:

1	2	3	4	5
0.19695	-0.22076	-0.11740	0.08273	0.05848

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.438e+00	1.529e-01	9.403	0.00255 **
normalized_stream_distance_m	-3.170e-04	8.088e-05	-3.919	0.02954 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1928 on 3 degrees of freedom

Multiple R-squared: 0.8366, Adjusted R-squared: 0.7822

F-statistic: 15.36 on 1 and 3 DF, p-value: 0.02954

```
# 7/23 - lc-excess & normalized stream distance
```

```
model.8 <- lm(`lc_excess_7_23` ~ normalized_stream_distance_m, data = gradient_it)  
summary(model.8)
```

Call:

```
lm(formula = lc_excess_7_23 ~ normalized_stream_distance_m, data = gradient_it)
```

Residuals:

1	2	3	4	5
0.18103	-0.20000	-0.05687	-0.04930	0.12514

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.005e+00	1.405e-01	7.154	0.00563 **
normalized_stream_distance_m	-2.609e-04	7.428e-05	-3.512	0.03914 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1771 on 3 degrees of freedom

Multiple R-squared: 0.8044, Adjusted R-squared: 0.7391

F-statistic: 12.33 on 1 and 3 DF, p-value: 0.03914

```
# Trail Creek
# lc-excess & cumulative dams

gradient_tc$cumulative_dams <- as.numeric(as.character(gradient_tc$cumulative_dams))

# 6/24 - lc-excess & cumulative dams
model.9 <- lm(`lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
summary(model.9)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.4806	0.1654	0.1151	0.4332	0.4826	-0.7157

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.613391	0.362589	-4.450	0.01125 *
cumulative_dams	-0.034669	0.006788	-5.107	0.00695 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5487 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.867, Adjusted R-squared: 0.8338

F-statistic: 26.08 on 1 and 4 DF, p-value: 0.006948

```
# 6/30 - lc-excess & cumulative dams
model.10 <- lm(`lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
summary(model.10)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
--	---	---	---	---	---	---

```
-0.27039 -0.01815  0.34655  0.43245 -0.47798 -0.01249
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.501209	0.257870	-9.699	0.000632 ***
cumulative_dams	-0.025481	0.004828	-5.278	0.006179 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3902 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8744, Adjusted R-squared: 0.843

F-statistic: 27.86 on 1 and 4 DF, p-value: 0.006179

```
# 7/15 - lc-excess & cumulative dams
model.11 <- lm(`lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
summary(model.11)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ cumulative_dams, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.55012	0.17680	0.52850	0.19513	-0.09976	-0.25056

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.241283	0.281127	-4.415	0.01155 *
cumulative_dams	-0.032606	0.005263	-6.195	0.00345 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4254 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9056, Adjusted R-squared: 0.882

F-statistic: 38.38 on 1 and 4 DF, p-value: 0.003452

```
# Coal Creek
# lc-excess & cumulative dams

gradient_cc$cumulative_dams <- as.numeric(as.character(gradient_cc$cumulative_dams))
```

```
# 6/23 - lc-excess & cumulative dams
model.12 <- lm(`lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
summary(model.12)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ cumulative_dams, data = gradient_cc)
```

Residuals:

2	3	4	5
0.11851	0.07466	-0.40553	0.21236

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.69377	0.41827	4.049	0.0559 .
cumulative_dams	-0.04687	0.01195	-3.921	0.0593 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3385 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8849, Adjusted R-squared: 0.8273

F-statistic: 15.37 on 1 and 2 DF, p-value: 0.05931

```
# 7/7 - lc-excess & cumulative dams
model.13 <- lm(`lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
summary(model.13)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ cumulative_dams, data = gradient_cc)
```

Residuals:

2	3	4	5
0.8361	-0.6905	-0.9591	0.8135

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.13831	1.45089	-1.474	0.278
cumulative_dams	-0.01660	0.04146	-0.400	0.728

```
Residual standard error: 1.174 on 2 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.0742, Adjusted R-squared: -0.3887
F-statistic: 0.1603 on 1 and 2 DF, p-value: 0.7276
```

```
# Trail Creek
# lc-excess & cumulative dam length
gradient_tc$cumulative_dam_length <- as.numeric(as.character(gradient_tc$cumulative_dam_length))

# 6/24 - lc-excess & cumulative dam length
model.14 <- lm(`lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
summary(model.14)
```

Call:

```
lm(formula = `lc-excess_6-24` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

```
      2      3      4      5      6      7
-0.3782  0.3121 -0.2486  0.3477  0.3552 -0.3882
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.715833	0.259857	-6.603	0.00273 **
cumulative_dam_length	-0.056153	0.008125	-6.911	0.00230 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Residual standard error: 0.4183 on 4 degrees of freedom
(1 observation deleted due to missingness)
```

```
Multiple R-squared: 0.9227, Adjusted R-squared: 0.9034
F-statistic: 47.77 on 1 and 4 DF, p-value: 0.002299
```

```
# 6/30 - lc-excess & cumulative dam length
model.15 <- lm(`lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
summary(model.15)
```

Call:

```
lm(formula = `lc-excess_6-30` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.16963	0.11233	0.09129	0.36273	-0.58878	0.19207

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.601968	0.233442	-11.15	0.000369 ***
cumulative_dam_length	-0.040214	0.007299	-5.51	0.005295 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3758 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.8836, Adjusted R-squared: 0.8545

F-statistic: 30.36 on 1 and 4 DF, p-value: 0.005295

```
# 7/15 - lc-excess & cumulative dam length
model.16 <- lm(`lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
summary(model.16)
```

Call:

```
lm(formula = `lc-excess_7-15` ~ cumulative_dam_length, data = gradient_tc)
```

Residuals:

	2	3	4	5	6	7
	-0.4429	0.3245	0.1916	0.1118	-0.2269	0.0420

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.348532	0.197391	-6.832	0.00240 **
cumulative_dam_length	-0.052360	0.006172	-8.484	0.00106 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3178 on 4 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9474, Adjusted R-squared: 0.9342

F-statistic: 71.97 on 1 and 4 DF, p-value: 0.001058

```
# Coal Creek
# lc-excess & cumulative dam length

gradient_cc$cumulative_dam_length <- as.numeric(as.character(gradient_cc$cumulative_dam_length))

# 6/23 - lc-excess & cumulative dam length
model.17 <- lm(`lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
summary(model.17)
```

Call:

```
lm(formula = `lc-excess_6-23` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

```
      2      3      4      5
0.10405 -0.06714 -0.20308  0.16617
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.547568	0.225772	6.855	0.0206 *
cumulative_dam_length	-0.027556	0.004094	-6.730	0.0214 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2052 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.9577, Adjusted R-squared: 0.9366

F-statistic: 45.29 on 1 and 2 DF, p-value: 0.02137

```
# 7/7 - lc-excess & cumulative dam length
model.18 <- lm(`lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
summary(model.18)
```

Call:

```
lm(formula = `lc-excess_7-7` ~ cumulative_dam_length, data = gradient_cc)
```

Residuals:

```
      2      3      4      5
0.7792 -0.7622 -0.8651  0.8481
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.11270	1.26808	-1.666	0.238
cumulative_dam_length	-0.01134	0.02300	-0.493	0.671

Residual standard error: 1.152 on 2 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.1083, Adjusted R-squared: -0.3375

F-statistic: 0.243 on 1 and 2 DF, p-value: 0.6709

```
# Trail Creek ISCO's
# Correlation test of discharge and lc-excess

# Downstream Trail Creek ISCO
cor.test(discharge_tc_isco$discharge, discharge_tc_isco$lc_excess)
```

Pearson's product-moment correlation

data: discharge_tc_isco\$discharge and discharge_tc_isco\$lc_excess

t = 24.555, df = 44, p-value < 2.2e-16

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.9379629 0.9808167

sample estimates:

cor

0.9653947

```
# Upstream Trail Creek ISCO
cor.test(discharge_utc_isco$discharge, discharge_utc_isco$lc_excess)
```

Pearson's product-moment correlation

data: discharge_utc_isco\$discharge and discharge_utc_isco\$lc_excess

t = 4.7379, df = 38, p-value = 3e-05

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.3676755 0.7739744

sample estimates:

cor

0.6093889


```
# Downstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_tc_isco)
```

Call:

```
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)
```

Coefficients:

(Intercept)	discharge
-5.986	42.209

```
# Upstream Trail Creek - lc-excess & discharge
lm(lc_excess ~ discharge, data = discharge_utc_isco)
```

Call:

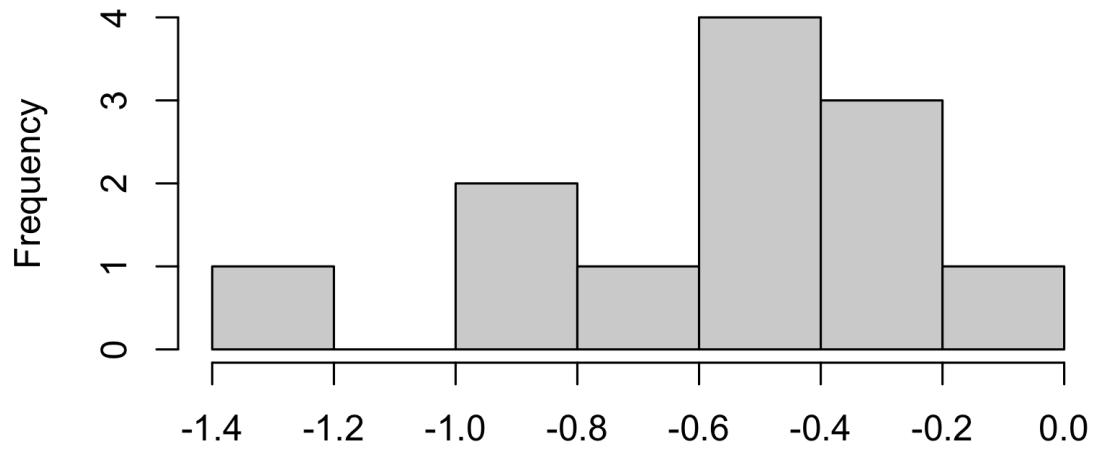
```
lm(formula = lc_excess ~ discharge, data = discharge_utc_isco)
```

Coefficients:

(Intercept)	discharge
-2.975	19.597

```
# Visual check
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]) # skewed left
```

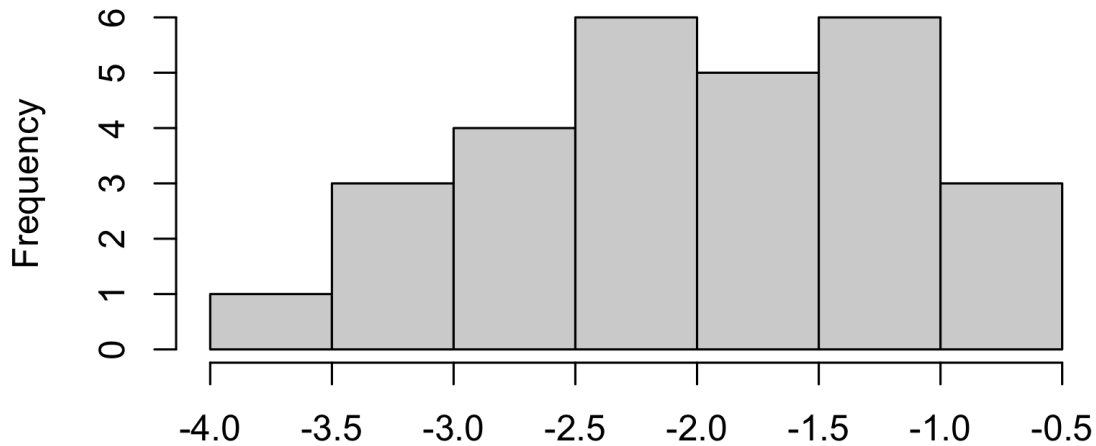
`f discharge_utc_isco$lc_excess[discharge_utc_isco$period`



`discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"`

```
hist(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]) # normal distrib
```

discharge_utc_isco\$lc_excess[discharge_utc_isco\$period



discharge_utc_isco\$lc_excess[discharge_utc_isco\$period == "post-pea

```
# Formal test (Shapiro-Wilk) - only reliable with reasonable sample size  
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"])
```

Shapiro-Wilk normality test

```
data: discharge_utc_isco$lc_excess[discharge_utc_isco$period == "pre-peak"]  
W = 0.97691, p-value = 0.9683
```

```
shapiro.test(discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"])
```

Shapiro-Wilk normality test

```
data: discharge_utc_isco$lc_excess[discharge_utc_isco$period == "post-peak"]  
W = 0.96427, p-value = 0.4377
```

```
# Parametric alternative (Welch's t-test, doesn't assume equal variance) - use if normal
t.test(lc_excess ~ period, data = discharge_utc_isco)
```

Welch Two Sample t-test

```
data:  lc_excess by period
t = -8.1949, df = 37.963, p-value = 6.38e-10
alternative hypothesis: true difference in means between group post-peak and group pre-
peak is not equal to 0
95 percent confidence interval:
 -1.784032 -1.077199
sample estimates:
mean in group post-peak  mean in group pre-peak
      -1.9979821          -0.5673667
```

```
discharge_utc_isco |>
  group_by(period) |>
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE)
  )
```

```
# A tibble: 2 x 5
  period      n mean_lc median_lc sd_lc
  <chr>    <int>   <dbl>    <dbl> <dbl>
1 post-peak    28   -2.00    -2.02  0.771
2 pre-peak     12   -0.567   -0.542  0.333
```

```
var.test(lc_excess ~ period, data = discharge_utc_isco)
```

F test to compare two variances

```
data:  lc_excess by period
F = 5.3505, num df = 27, denom df = 11, p-value = 0.005712
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
```

```

1.702796 13.452718
sample estimates:
ratio of variances
      5.350494

```

```

# Correlation of the lc-excess & location

compiled_isotope$date_c <- as.numeric(compiled_isotope$date) - mean(as.numeric(compiled_isotope$date))
# center the date variable by subtracting its mean.

model_location <- lmer(lc_excess ~ location + date_c + (1 | site), data = compiled_isotope)
# fixed effects: location differences + linear time trend
# random intercept: allows each site to have its own baseline lc-excess, accounting for
# the ~36% of variance that's due to site-level clustering rather than location or date
summary(model_location)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: lc_excess ~ location + date_c + (1 | site)
Data: compiled_isotope

```

```

REML criterion at convergence: 334.7

```

```

Scaled residuals:
      Min       1Q   Median       3Q      Max
-2.18174 -0.59314 -0.05398  0.65138  2.16327

```

```

Random effects:
Groups   Name             Variance Std.Dev.
site     (Intercept)  0.4233   0.6506
Residual                    0.8648   0.9300
Number of obs: 115, groups: site, 20

```

```

Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)   0.28574    0.38862 25.19091   0.735   0.4690
locationit     2.91690    0.53729 19.55019   5.429 2.79e-05 ***
locationntc    -2.88718    0.52093 16.75436  -5.542 3.77e-05 ***
locationuttc   -1.55173    0.54866 15.90779  -2.828   0.0122 *
date_c         -0.04662    0.00458 109.92129 -10.179 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation of Fixed Effects:

```
(Intr) loctnt locationtc locationutc
locationit -0.656
locationtc -0.722 0.501
locationutc -0.692 0.473 0.506
date_c      -0.236 -0.114 0.073      0.099
```

```
# centering doesn't affect slopes, only the intercept's ^ from the (1|site)
anova(model_location)
```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
location	111.828	37.276	3	15.082	43.102	1.238e-07 ***
date_c	89.602	89.602	1	109.921	103.605	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# the enrichment effect consistent across geologically distinct sites? This is a reach × creek
```

```
model_reach <- lmer(lc_excess ~ reach * creek + (1 | date), data = us_ds)
summary(model_reach)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

lmerModLmerTest]

Formula: lc_excess ~ reach * creek + (1 | date)

Data: us_ds

REML criterion at convergence: 27.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.05323	-0.31166	-0.03435	0.35767	0.96783

Random effects:

Groups	Name	Variance	Std.Dev.
date	(Intercept)	1.19749	1.0943
Residual		0.09793	0.3129

Number of obs: 16, groups: date, 8

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-1.3596	0.8048	5.3922	-1.689	0.14768
reachupstream	1.9010	0.3129	5.0000	6.075	0.00175
creekItalian Creek	2.7533	1.0390	5.3922	2.650	0.04217
creekTrail Creek	-3.3642	1.0390	5.3922	-3.238	0.02064
reachupstream:creekItalian Creek	-1.1078	0.4040	5.0000	-2.742	0.04069
reachupstream:creekTrail Creek	1.0219	0.4040	5.0000	2.530	0.05256

```

(Intercept)
reachupstream      **
creekItalian Creek *
creekTrail Creek   *
reachupstream:creekItalian Creek *
reachupstream:creekTrail Creek .

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

(Intr) rchpst crkItC crkTrC rch:IC
reachupstrm -0.194
crekItlnCrk -0.775 0.151
creekTrlCrk -0.775 0.151 0.600
rchpstrm:IC 0.151 -0.775 -0.194 -0.117
rchpstrm:TC 0.151 -0.775 -0.117 -0.194 0.600

```

`anova(model_reach)`

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
reach	13.5224	13.5224	1	5	138.082	7.849e-05 ***
creek	3.0212	1.5106	2	5	15.425	0.007264 **
reach:creek	3.4032	1.7016	2	5	17.376	0.005611 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

the lmer() model is more powerful here because it pools information across all three creeks

What the model is actually telling you:

`reach` main effect (p = 7.8e-05): across all sites, upstream vs. downstream location matters

`creek` main effect (p = 0.0073): the three creeks differ in baseline lc-excess overall, with Italian Creek having the highest and Trail Creek the lowest

```
# `reach:creek` interaction (p = 0.0056): this is the key term for aim (ii). It's significant
# Looking at the coefficients:
# Italian Creek's interaction term is -1.11 (p = 0.041) - the upstream/downstream gap is small
# Italian Creek is your undammed control, so a weaker reach effect there strengthens your case
# Trail Creek's interaction term is +1.02 (p = 0.053, right at the edge) - trending toward a
```

```
lc_stat <- river_reach |>
  group_by(location) |>
  summarise(
    n = n(),
    mean_lc = mean(lc_excess, na.rm = TRUE),
    sd_lc = sd(lc_excess, na.rm = TRUE),
    median_lc = median(lc_excess, na.rm = TRUE)
  ) |>
  mutate(across(where(is.numeric), ~ round(.x, 3)))

table7 <- lc_stat |>
  rename(
    Location = location,
    N = n,
    `Mean (%)` = mean_lc,
    `SD (%)` = sd_lc,
    `Median (%)` = median_lc
  ) |>
  gt() |>
  tab_header(
    title = "Table 7. Descriptive statistics of lc-excess by sampling location"
  ) |>
  cols_align(align = "center", columns = everything()) |>
  cols_align(align = "left", columns = Location) |>
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels()
  ) |>
  tab_options(
    table.font.size = px(13),
    heading.title.font.size = px(13),
    column_labels.background.color = "#f2f2f2"
  )
```



```

# =====
# Compile all statistical tests into publication-ready tables
# =====

# -----
# TABLE 1. Pairwise site comparisons (ANOVA + Tukey HSD)
# -----

tukey_tidy <- tidy(TukeyHSD(anova)) %>%
  select(comparison = contrast, diff = estimate,
         lower = conf.low, upper = conf.high, p_adj = adj.p.value)

aov_summary <- summary(anova)[[1]]

f_stat <- aov_summary$`F value`[1]
df1    <- aov_summary$Df[1]
df2    <- aov_summary$Df[2]
p_val  <- aov_summary$`Pr(>F)`[1]

table1 <- tukey_tidy %>%
  gt() %>%
  tab_header(
    title = "Table 1. Pairwise comparisons of lc-excess among sampling locations",
    subtitle = "Tukey HSD following one-way ANOVA"
  ) %>%
  fmt_number(columns = c(diff, lower, upper), decimals = 2) %>%
  fmt_number(columns = p_adj, decimals = 4) %>%
  cols_label(
    comparison = "Comparison", diff = "Mean diff.",
    lower = "95% CI (lower)", upper = "95% CI (upper)", p_adj = "Adj. p"
  ) %>%
  gt::tab_source_note(
    source_note = paste0(
      "Overall ANOVA: F(", df1, ", ", df2, ") = ", round(f_stat, 2), ", p < 0.001"
    )
  )

table1

# -----
# TABLE 2. Mixed model: lc-excess ~ location + date (site random effect)
# -----

```

Table 1. Pairwise comparisons of lc-excess among sampling locations
Tukey HSD following one-way ANOVA

Comparison	Mean diff.	95% CI (lower)	95% CI (upper)	Adj. p
it-cc	2.30	0.90	3.69	0.0002
tc-cc	-1.90	-3.06	-0.74	0.0003
utc-cc	-0.75	-1.94	0.43	0.3520
tc-it	-4.19	-5.26	-3.13	0.0000
utc-it	-3.05	-4.14	-1.96	0.0000
utc-tc	1.15	0.37	1.92	0.0010

Overall ANOVA: $F(3,111) = 36.53$, $p < 0.001$

```
fixed_loc <- tidy(model_location, effects = "fixed") %>%
  select(term, estimate, std.error, df, statistic, p.value)

rand_loc <- tidy(model_location, effects = "ran_pars") %>%
  select(group, term, estimate)

table2 <- fixed_loc %>%
  gt() %>%
  tab_header(
    title = "Table 2. Linear mixed-effects model of lc-excess",
    subtitle = "Fixed effects: sampling location and date; random intercept: site"
  ) %>%
  fmt_number(columns = c(estimate, std.error, df, statistic), decimals = 3) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(
    term = "Term", estimate = "Estimate", std.error = "SE",
    df = "df", statistic = "t", p.value = "p"
  ) %>%
  tab_source_note(
    source_note = paste0(
      "Random effect variance (site) = ",
      round(rand_loc$estimate[rand_loc$group == "site"], 3),
      "; N = ", nrow(compiled_isotope), " observations, ",
      n_distinct(compiled_isotope$site), " sites"
    )
  )
```

Table 2. Linear mixed-effects model of lc-excess
Fixed effects: sampling location and date; random intercept: site

Term	Estimate	SE	df	t	p
(Intercept)	0.286	0.389	25.191	0.735	0.4690
locationit	2.917	0.537	19.550	5.429	0.0000
locationtc	-2.887	0.521	16.754	-5.542	0.0000
locationutc	-1.552	0.549	15.908	-2.828	0.0122
date_c	-0.047	0.005	109.921	-10.179	0.0000

Random effect variance (site) = 0.651; N = 115 observations, 20 sites

table2

```
# -----
# TABLE 3. Spatial gradient regressions (lc-excess vs. distance /
# cumulative dams / cumulative dam length), all creeks & dates
# -----

gradient_models <- list(
  list(model = model.1, creek = "Trail Creek", date = "6/24", predictor = "Stream distance"),
  list(model = model.2, creek = "Trail Creek", date = "6/30", predictor = "Stream distance"),
  list(model = model.3, creek = "Trail Creek", date = "7/15", predictor = "Stream distance"),
  list(model = model.4, creek = "Coal Creek", date = "6/23", predictor = "Stream distance"),
  list(model = model.5, creek = "Coal Creek", date = "7/7", predictor = "Stream distance"),
  list(model = model.6, creek = "Italian Creek", date = "7/1", predictor = "Stream distance"),
  list(model = model.7, creek = "Italian Creek", date = "7/16", predictor = "Stream distance"),
  list(model = model.8, creek = "Italian Creek", date = "7/23", predictor = "Stream distance"),
  list(model = model.9, creek = "Trail Creek", date = "6/24", predictor = "Cumulative dams"),
  list(model = model.10, creek = "Trail Creek", date = "6/30", predictor = "Cumulative dams"),
  list(model = model.11, creek = "Trail Creek", date = "7/15", predictor = "Cumulative dams"),
  list(model = model.12, creek = "Coal Creek", date = "6/23", predictor = "Cumulative dams"),
  list(model = model.13, creek = "Coal Creek", date = "7/7", predictor = "Cumulative dams"),
  list(model = model.14, creek = "Trail Creek", date = "6/24", predictor = "Cumulative dam"),
  list(model = model.15, creek = "Trail Creek", date = "6/30", predictor = "Cumulative dam"),
  list(model = model.16, creek = "Trail Creek", date = "7/15", predictor = "Cumulative dam"),
  list(model = model.17, creek = "Coal Creek", date = "6/23", predictor = "Cumulative dam"),
  list(model = model.18, creek = "Coal Creek", date = "7/7", predictor = "Cumulative dam")
)
```

```

extract_lm <- function(entry) {
  m <- entry$model
  s <- summary(m)
  co <- coef(s)
  tibble(
    Predictor = entry$predictor,
    Creek     = entry$creek,
    Date      = entry$date,
    Intercept = co[1, 1],
    Slope     = co[2, 1],
    SE        = co[2, 2],
    R2        = s$r.squared,
    p_value   = co[2, 4]
  )
}

table3_df <- map_dfr(gradient_models, extract_lm)

table3 <- table3_df %>%
  gt(groupname_col = "Predictor", rowname_col = "Creek") %>%
  tab_header(
    title = "Table 3. Spatial gradient regressions of lc-excess",
    subtitle = "lc-excess regressed against normalized stream distance, cumulative beaver dam
  ) %>%
  fmt_number(columns = c(Intercept, Slope, SE, R2), decimals = 3) %>%
  fmt_number(columns = p_value, decimals = 4) %>%
  cols_label(
    Date = "Date", Intercept = "Intercept", Slope = "Slope",
    SE = "SE (slope)", R2 = "R\u00b2", p_value = "p"
  ) %>%
  tab_options(row_group.as_column = FALSE)

table3

# -----
# TABLE 4. Discharge - lc-excess relationship, upstream vs downstream
# -----

corr_us <- tidy(cor.test(discharge_utc_isco$discharge, discharge_utc_isco$lc_excess)) %>%
  mutate(Reach = "Upstream")
corr_ds <- tidy(cor.test(discharge_tc_isco$discharge, discharge_tc_isco$lc_excess)) %>%
  mutate(Reach = "Downstream")

```

Table 3. Spatial gradient regressions of lc-excess
lc-excess regressed against normalized stream distance, cumulative beaver
dam count, and cumulative dam length, by creek and sampling date

	Date	Intercept	Slope	SE (slope)	R ²	p
Stream distance (m)						
Trail Creek	6/24	-1.391	-0.001	0.000	0.917	0.0007
Trail Creek	6/30	-2.280	-0.001	0.000	0.893	0.0013
Trail Creek	7/15	-1.103	-0.001	0.000	0.892	0.0014
Coal Creek	6/23	1.539	-0.001	0.000	0.740	0.0614
Coal Creek	7/7	-0.333	-0.001	0.000	0.652	0.0986
Italian Creek	7/1	3.549	0.000	0.000	0.278	0.3617
Italian Creek	7/16	1.438	0.000	0.000	0.837	0.0295
Italian Creek	7/23	1.005	0.000	0.000	0.804	0.0391
Cumulative dams (n)						
Trail Creek	6/24	-1.613	-0.035	0.007	0.867	0.0069
Trail Creek	6/30	-2.501	-0.025	0.005	0.874	0.0062
Trail Creek	7/15	-1.241	-0.033	0.005	0.906	0.0035
Coal Creek	6/23	1.694	-0.047	0.012	0.885	0.0593
Coal Creek	7/7	-2.138	-0.017	0.041	0.074	0.7276
Cumulative dam length (m)						
Trail Creek	6/24	-1.716	-0.056	0.008	0.923	0.0023
Trail Creek	6/30	-2.602	-0.040	0.007	0.884	0.0053
Trail Creek	7/15	-1.349	-0.052	0.006	0.947	0.0011
Coal Creek	6/23	1.548	-0.028	0.004	0.958	0.0214
Coal Creek	7/7	-2.113	-0.011	0.023	0.108	0.6709

Table 4. Relationship between discharge and lc-excess
Trail Creek ISCO stations, Pearson correlation and linear regression slope

Reach	Pearson r	95% CI (lower)	95% CI (upper)	p	n	Regression slope
Upstream	0.609	0.368	0.774	0.0000	40	19.597
Downstream	0.965	0.938	0.981	0.0000	46	42.209

```
lm_us <- tidy(lm(lc_excess ~ discharge, data = discharge_utc_isco)) %>%
  filter(term == "discharge") %>% mutate(Reach = "Upstream")
lm_ds <- tidy(lm(lc_excess ~ discharge, data = discharge_tc_isco)) %>%
  filter(term == "discharge") %>% mutate(Reach = "Downstream")

table4_df <- bind_rows(corr_us, corr_ds) %>%
  select(Reach, r = estimate, conf.low, conf.high, corr_p = p.value, n = parameter) %>%
  mutate(n = n + 2) %>% # cor.test df = n - 2
  left_join(
    bind_rows(lm_us, lm_ds) %>% select(Reach, slope = estimate),
    by = "Reach"
  )

table4 <- table4_df %>%
  gt() %>%
  tab_header(
    title = "Table 4. Relationship between discharge and lc-excess",
    subtitle = "Trail Creek ISCO stations, Pearson correlation and linear regression slope"
  ) %>%
  fmt_number(columns = c(r, conf.low, conf.high, slope), decimals = 3) %>%
  fmt_number(columns = corr_p, decimals = 4) %>%
  cols_label(
    Reach = "Reach", r = "Pearson r", conf.low = "95% CI (lower)",
    conf.high = "95% CI (upper)", corr_p = "p", n = "n", slope = "Regression slope"
  )
table4
```

```
# -----
# TABLE 5. Pre- vs. post-peak flow lc-excess comparison
# -----

period_summary <- discharge_isco %>%
  group_by(period) %>%
```

```

summarise(
  n = n(), mean = mean(lc_excess, na.rm = TRUE),
  sd = sd(lc_excess, na.rm = TRUE), median = median(lc_excess, na.rm = TRUE),
  .groups = "drop"
)

test_results <- bind_rows(
  tidy(t.test(lc_excess ~ period, data = discharge_isco)) %>%
    mutate(Test = "Welch two-sample t-test") %>%
    select(Test, statistic, df = parameter, p.value),
  tidy(wilcox.test(lc_excess ~ period, data = discharge_utc_isco)) %>%
    mutate(Test = "Mann-Whitney U", df = NA_real_) %>%
    select(Test, statistic, df, p.value),
  tidy(var.test(lc_excess ~ period, data = discharge_utc_isco)) %>%
    mutate(Test = "F test (equality of variance)") %>%
    select(Test, statistic, df = num.df, p.value)
)

```

Multiple parameters; naming those columns num.df and den.df.

```

table5a <- period_summary %>%
  gt() %>%
  tab_header(title = "Table 5a. lc-excess summary statistics by flow period",
             subtitle = "Upstream and Downstream Trail Creek ISCO (peak flow: 5/14/2026)") %>%
  fmt_number(columns = c(mean, sd, median), decimals = 3) %>%
  cols_label(period = "Period", n = "n", mean = "Mean", sd = "SD", median = "Median")

table5b <- test_results %>%
  gt() %>%
  tab_header(title = "Table 5b. Statistical tests, pre- vs. post-peak lc-excess") %>%
  fmt_number(columns = c(statistic, df), decimals = 2) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(Test = "Test", statistic = "Statistic", df = "df", p.value = "p")

table5a

```

```

table5b

```

Table 5a. lc-excess summary statistics by flow period
Upstream and Downstream Trail Creek ISCO (peak flow: 5/14/2026)

Period	n	Mean	SD	Median
post-peak	62	-2.980	1.739	-2.458
pre-peak	24	-0.727	0.352	-0.734

Table 5b. Statistical tests, pre- vs. post-peak lc-excess

Test	Statistic	df	p
Welch two-sample t-test	-9.70	72.46	0.0000
Mann-Whitney U	9.00	NA	0.0000
F test (equality of variance)	5.35	27.00	0.0057

```
# -----
# TABLE 6. Upstream/downstream mixed model (reach x creek)
# -----

fixed_reach <- tidy(model_reach, effects = "fixed") %>%
  select(term, estimate, std.error, df, statistic, p.value)

anova_reach <- tidy(anova(model_reach)) %>%
  select(term, NumDF = NumDF, DenDF = DenDF, statistic, p.value)
```

Warning: The column names NumDF and DenDF in ANOVA output were not recognized or transformed.

```
table6a <- fixed_reach %>%
  gt() %>%
  tab_header(title = "Table 6a. Fixed effects: lc-excess ~ reach * creek",
             subtitle = "Random intercept: sampling date") %>%
  fmt_number(columns = c(estimate, std.error, df, statistic), decimals = 3) %>%
  fmt_number(columns = p.value, decimals = 4) %>%
  cols_label(term = "Term", estimate = "Estimate", std.error = "SE",
             df = "df", statistic = "t", p.value = "p")

table6b <- anova_reach %>%
  gt() %>%
  tab_header(title = "Table 6b. Type III ANOVA, lc-excess ~ reach * creek") %>%
```


Table 6a. Fixed effects: lc-excess ~ reach * creek
Random intercept: sampling date

Term	Estimate	SE	df	t	p
(Intercept)	-1.360	0.805	5.392	-1.689	0.1477
reachupstream	1.901	0.313	5.000	6.075	0.0017
creekItalian Creek	2.753	1.039	5.392	2.650	0.0422
creekTrail Creek	-3.364	1.039	5.392	-3.238	0.0206
reachupstream:creekItalian Creek	-1.108	0.404	5.000	-2.742	0.0407
reachupstream:creekTrail Creek	1.022	0.404	5.000	2.530	0.0526

Table 6b. Type III ANOVA, lc-excess ~ reach * creek

Term	Num. df	Den. df	F	p
reach	1.00	5.00	138.08	0.0001
creek	2.00	5.00	15.43	0.0073
reach:creek	2.00	5.00	17.38	0.0056

```
fmt_number(columns = c(NumDF, DenDF, statistic), decimals = 2) %>%
fmt_number(columns = p.value, decimals = 4) %>%
cols_label(term = "Term", NumDF = "Num. df", DenDF = "Den. df",
            statistic = "F", p.value = "p")

table6a
```

```
table6b
```

```
table7 <- lc_stat |>
  rename(Location = location, N = n, `Mean (%)` = mean_lc, `SD (%)` = sd_lc, `Median (%)` = median_lc) |>
  gt() |>
  tab_header(
    title = "Table 7. Descriptive statistics of lc-excess by sampling location") |>
  cols_align(aligned = "center", columns = everything()) |>
  cols_align(aligned = "left", columns = Location)

table7
```

Table 7. Descriptive statistics of lc-excess by sampling location

Location	N	Mean (‰)	SD (‰)	Median (‰)
cc	12	-0.647	1.669	-0.262
it	15	1.648	1.330	0.966
tc	12	-3.513	1.106	-3.410
utc	9	-1.847	0.676	-1.825

```
## Table 8
q_paired <- sontek %>%
  group_by(creek, date) %>%
  filter(site_num == min(site_num) | site_num == max(site_num)) %>%
  mutate(position = ifelse(site_num == min(site_num), "upstream", "downstream")) %>%
  ungroup() %>%
  select(date, creek, position, q_cfs) %>%
  distinct(date, creek, position, .keep_all = TRUE) %>% # in case of ties
  pivot_wider(names_from = position, values_from = q_cfs) %>%
  mutate(
    diff_cfs = downstream - upstream,
    pct_change = (downstream - upstream) / upstream * 100
  ) %>%
  drop_na(upstream, downstream)

q_paired

# A tibble: 5 x 6
  date      creek upstream downstream diff_cfs pct_change
<chr>   <chr>   <dbl>     <dbl>    <dbl>     <dbl>
1 6/23/26 CC      1.35      1.88     0.525     38.9
2 7/1/26  IT      2.85      4.49     1.64      57.7
3 7/7/26  CC      0.069     0.232    0.163     236.
4 7/16/26 IT      3.07      3.41     0.34      11.1
5 7/23/26 IT      2.89      4.72     1.84      63.6

# gt table for the paper
table8 <- q_paired |>
  arrange(creek, date) |>
  gt() |>
  fmt_number(columns = c(upstream, downstream, diff_cfs, pct_change), decimals = 3) |>
```

Table 8. Paired discharge comparison, upstream vs. downstream sites by creek

Date	Creek	Upstream Q (cfs)	Downstream Q (cfs)	Difference (cfs)	% Change
6/23/26	CC	1.351	1.876	0.525	38.860
7/7/26	CC	0.069	0.232	0.163	236.232
7/1/26	IT	2.849	4.492	1.643	57.669
7/16/26	IT	3.072	3.412	0.340	11.068
7/23/26	IT	2.886	4.721	1.835	63.583

```
cols_label(
  date = "Date",
  creek = "Creek",
  upstream = "Upstream Q (cfs)",
  downstream = "Downstream Q (cfs)",
  diff_cfs = "Difference (cfs)",
  pct_change = "% Change"
) |>
tab_header(title = "Table 8. Paired discharge comparison, upstream vs. downstream sites by
table8
```

```
# -----
# Save all tables (as individual Word-ready files)
# -----

dir.create("tables", showWarnings = FALSE)
gtsave(table1, "tables/table1_tukey.docx")
gtsave(table2, "tables/table2_location_model.docx")
gtsave(table3, "tables/table3_gradient_models.docx")
gtsave(table4, "tables/table4_discharge.docx")
gtsave(table5a, "tables/table5a_period_summary.docx")
gtsave(table5b, "tables/table5b_period_tests.docx")
gtsave(table6a, "tables/table6a_reach_creek_fixed.docx")
gtsave(table6b, "tables/table6b_reach_creek_anova.docx")
gtsave(table7, "tables/table7_statistics_river_reach.docx")
gtsave(table8, "tables/table8_discharge_cc_it.docx")
```